

02

Web Application Development

Modern and scalable web applications for businesses, organizations and digital platforms.

Project Overview

A service organisation needed to move a slow, ageing internal portal to a modern web application that could handle growing traffic, support multiple user roles and remain maintainable as the team expanded.

The Challenge

- The legacy portal loaded slowly and struggled under concurrent users.
- Business logic was scattered through the codebase, making changes risky.
- The interface was not usable on mobile devices, though most staff worked from phones.
- There was no structured deployment process, so releases often broke production.

Our Solution

- Rebuilt the application on Laravel using a clean, modular structure with reusable services.
- Built responsive interfaces with Bootstrap and Vue.js components for dynamic screens.
- Introduced query optimisation, eager loading and caching to remove database bottlenecks.
- Exposed a documented REST API so future mobile and partner integrations are straightforward.
- Set up Git-based version control with branch-per-feature and reviewed pull requests on GitHub.
- Added form validation, error handling and monitoring for safer releases.

Technologies Used

Frontend	HTML5, CSS3, JavaScript, Bootstrap, Vue.js
Backend	PHP, Laravel, REST API
Database	MySQL
Dev & Collaboration	Git, GitHub, VS Code, XAMPP

Results

Metric	Before	After
Average page load time	approx. 4.5 s	approx. 1.2 s
Mobile usability	Not supported	Fully responsive
Concurrent users handled	Limited	Comfortably scaled
Release failures	Common	Rare, with rollback path

Key Takeaway

Scalability is mostly an architecture decision, not a hosting decision. A clean Laravel structure, a well-designed database and a disciplined Git workflow keep an application fast and safe to change long after launch.

Note: Figures shown are representative project outcomes and can be replaced with your own verified client data before sharing.